

Fine-tuning SBERT for Domain-Specific Sentence Similarity in Healthcare

Alessandro Quercetti

Department of
Computer Science
Reykjavik University,
Reykjavik, Iceland
alessandro25@ru.is

Damiano Buzzo

Department of
Computer Science
Reykjavik University,
Reykjavik, Iceland
damiano25@ru.is

Samuele Pirani

Department of
Computer Science
Reykjavik University,
Reykjavik, Iceland
samuele25@ru.is

Abstract

This study investigates domain adaptation of Sentence-BERT (SBERT) for semantic similarity, clustering, and retrieval of clinical texts. Using OHSUMED abstracts and MeSH-based weak supervision, the model is fine-tuned with contrastive learning to better capture biomedical topical structure. Representation quality is evaluated through human-aligned similarity, PCA-based clustering, and dense-only retrieval. Results show that fine-tuning improves global embedding coherence and yields modest gains in deeper-rank retrieval performance, while producing small decreases in correlation with human similarity judgments. These findings suggest that MeSH-guided adaptation enhances topical organization in the embedding space and provides a strong foundation for downstream clinical information retrieval pipelines.

1 Introduction

The increasing volume of unstructured text data in healthcare, particularly clinical notes and biomedical literature, necessitates advanced Natural Language Processing (NLP) techniques for efficient management and information retrieval (IR). Extracting actionable information from clinical text is challenging due to language variability, inconsistent use of medical terminology, and lack of standardized formatting (Bhopale and Tiwari, 2024) (Kumar et al., 2025). Transformer-based architectures, such as Bidirectional Encoder Representations from Transformers (BERT), have revolutionized this field by providing contextual representations of text. These models generate embedding vectors based on the surrounding context, effectively addressing polysemy. However, applying generalist models directly to clinical text faces challenges, partly because of the domain’s context-heavy nature (Myers et al., 2025). General-context BERT or Sentence-BERT (SBERT) models, while efficient, may not comprehensively cover

specialized clinical knowledge and medical meanings without tailored training. For instance, studies show that lexical methods like Rank BM25 can consistently outperform general SBERT models in incident report search tasks (Zoie, 2024). Moreover, transformer models suffer from limitations concerning the maximum input length (often restricted to 512 tokens), which makes processing lengthy clinical documents difficult. The efficiency of traditional BERT is also constrained by long running times when assessing semantic similarity (Ortakci, 2024), due to its cross-encoder architecture, which performs full self-attention over the inputs. This high computational cost prompted the development of efficient variants.

SBERT was developed to overcome BERT’s inefficiency in semantic search by using a Siamese network architecture to generate fixed-size vector representations (sentence embeddings) efficiently. This capability makes SBERT highly suitable for document similarity tasks, including large-scale clustering (Ortakci, 2024). This study aims to address the domain-specific challenges in healthcare by developing and evaluating a transformer-based framework centered on fine-tuning SBERT for two main applications in the clinical domain: clustering and retrieval. We hypothesize that integrating techniques designed to enhance contextual representation and handle practical constraints will significantly boost performance over generalist methods. We investigate the following research questions:

- RQ1 – Domain adaptation: How much does fine-tuning SBERT on healthcare-specific sentence pairs improve sentence similarity performance compared to general-domain SBERT models?
- RQ2 – Clustering effectiveness: How effectively do sentence embeddings from fine-tuned SBERT capture semantic structure in

healthcare texts for clustering purposes, compared to general-domain embeddings?

- RQ3 – Retrieval performance: Does using fine-tuned SBERT embeddings improve the retrieval of semantically related documents compared to general-domain embeddings?

2 Related Works

2.1 Evolution of Text Representation and SBERT

Classical IR methods such as the Vector Space Model, TF-IDF, and Okapi BM25 are efficient for lexical matching but cannot model semantic relationships (Zoie, 2024). Early neural approaches (e.g., Word2Vec) introduced distributed representations but produced static, context-independent embeddings for polysemous terms (Wang and Sun, 2022). The introduction of BERT enabled contextual encoding, substantially improving semantic retrieval, though its computational cost limits large-scale similarity scoring (Ortakci, 2024). Moreover, BERT does not naturally generate independent sentence embeddings suitable for direct comparison.

SBERT overcomes these limitations by using a Siamese architecture to compute fixed-size, semantically meaningful embeddings efficiently (Reimers and Gurevych, 2019). The architecture typically applies a transformer encoder (e.g., BERT-base) followed by a pooling operation (commonly mean pooling) to obtain sentence-level representations. Its objective is to map semantically similar texts close together in vector space, which makes SBERT a strong backbone for retrieval and clustering applications.

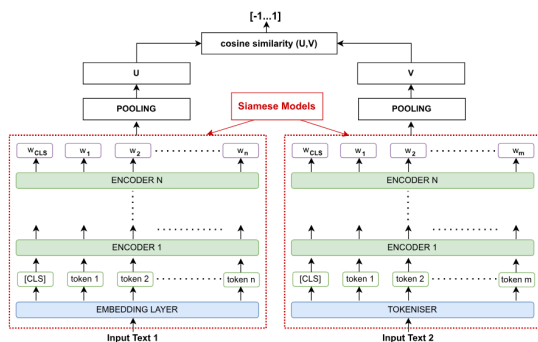


Figure 1: SBERT architecture (Ortakci, 2024)

2.2 Challenges and Domain Adaptation in Healthcare

Applying general LLMs to clinical text is challenging due to domain-specific terminology and context dependence. Models such as SBERT trained on general corpora underperform on biomedical retrieval tasks because they lack coverage of clinical concepts (Shi et al., 2022). Domain-specific encoders such as BioBERT and ClinicalBERT were introduced to address these limitations through pre-training on biomedical literature (Yang et al., 2024). Further improvements rely on self-supervised fine-tuning strategies, including contrastive learning (e.g., SimCSE) (Wang and Sun, 2022) and TSDAE (Kumar et al., 2025), to adapt generalist models to long-context clinical applications. These approaches aim to refine embeddings to better capture medical semantics and mitigate issues of polysemy pervasive in clinical narratives.

2.3 Key Tasks for SBERT Models

SBERT embeddings support various downstream tasks in healthcare NLP, particularly retrieval and clustering.

Retrieval: Clinical retrieval pipelines often combine summarization and embedding-based retrieval with cross-encoder re-ranking (Bhopale and Tiwari, 2024). Summarization mitigates BERT’s input length constraints, while embedding-based query expansion addresses the semantic gap in short clinical queries. Fine-tuned SBERT encoders in two-tower architectures efficiently generate document and query embeddings, with cross-encoders used for final ranking when higher accuracy is required.

Clustering: SBERT is effective for clustering due to its dense, low-dimensional embeddings (Ortakci, 2024). Integrations such as kBERT improve topic modeling in short clinical feedback (Islam et al., 2025), while clustering biomedical eligibility criteria using fine-tuned embeddings achieves high semantic fidelity (Bornet et al., 2025). Determining the optimal number of clusters remains a challenge; hybrid methods combining dimensionality reduction and optimization techniques (e.g., PCA+PSO) have been proposed (Ortakci, 2024). More advanced systems, including SBERT-BERT hybrids and GNN-based frameworks like HAMLET (Sakai and Lam, 2025), further enhance topic coherence and interpretability.

3 Implementation

This section describes the implemented pipeline¹ from the collection of data, preprocessing and for fine-tuning, evaluation, clustering, and information retrieval in the clinical domain. The figure 2 shows the pipeline used to implement our system.

3.1 Dataset and Preprocessing

The **OHSUMED** collection, derived from MEDLINE abstracts annotated with **MeSH** (Medical Subject Headings) terms, was employed for this study. The original train/test split was retained. For each record, the title, abstract, MeSH terms (serving as labels), and unique identifiers were preserved. The preprocessing phase included:

- Removal of records with missing or empty abstracts.
- Elimination of duplicate abstracts.
- Normalization of MeSH strings by extracting common descriptors (the term segment preceding “/”).

After preprocessing, the training and test sets were reduced from approximately 290k and 50k documents to 190k and 35k, respectively representing about a 30% reduction.

MeSH terms play a central role in the experimental design:

1. Providing weak supervision for mining semantically related (positive) pairs.
2. Serving as proxy class labels for evaluating clustering quality.
3. Acting as relevance indicators for intrinsic retrieval evaluation.

3.2 Fine-tuning

All downstream analyses rely on dense, sentence-level embeddings generated by a **Sentence-BERT (SBERT)** encoder implemented through the *sentence_transformers* library. The experimental pipeline consists of two main stages:

- **Baseline representations:** The test corpus was first encoded using the unmodified *all-MiniLM-L6-v2* checkpoint to establish a baseline semantic space. Cosine similarity was employed as the principal distance metric.

- **Domain adaptation:** To specialize the encoder for biomedical semantics, the same SBERT backbone was fine-tuned using **contrastive learning** with the *Multiple Negatives Ranking Loss*. Positive pairs were automatically mined from the training split by identifying abstracts that shared at least one MeSH term, while in-batch samples served as implicit hard negatives. A held-out subset of manually constructed positive and negative pairs was evaluated using an *Embedding Similarity Evaluator* to select the optimal checkpoint. The fine-tuned model was then applied to re-encode the test corpus, yielding two directly comparable embedding spaces—baseline and domain-adapted—for subsequent quantitative analyses.

In preliminary experiments, several fine-tuning runs were conducted with different data configurations. An initial setup inverted the train and test partitions. However, due to the substantial imbalance in their sizes, this configuration was abandoned in favor of the last used split. The final training run, which used the cleaned training portion for contrastive fine-tuning and the balanced validation subset for checkpoint selection, required slightly more than two hours on Google Colab. Longer runs were attempted, but Colab’s session time limits prevented their completion. The final completed run produced better results and was therefore adopted as the definitive fine-tuned model.

3.3 Representation evaluation

Representation quality is evaluated through (i) human-aligned similarity with baseline and fine-tuned embeddings compared under identical conditions.

For human-aligned similarity, 300 abstract pairs are constructed across three strata:

- *MeSH-overlap pairs*, sharing at least one MeSH term.
- *Nearest-neighbor pairs*, identified in the baseline space.
- *Random pairs*, serving as control group.

Human annotators assign similarity scores to each pair. Pearson and Spearman correlation coefficients are computed between model cosine similarities and the mean human ratings. In addition, a

¹<https://github.com/atomicSteiner/HealthcareSBERT>

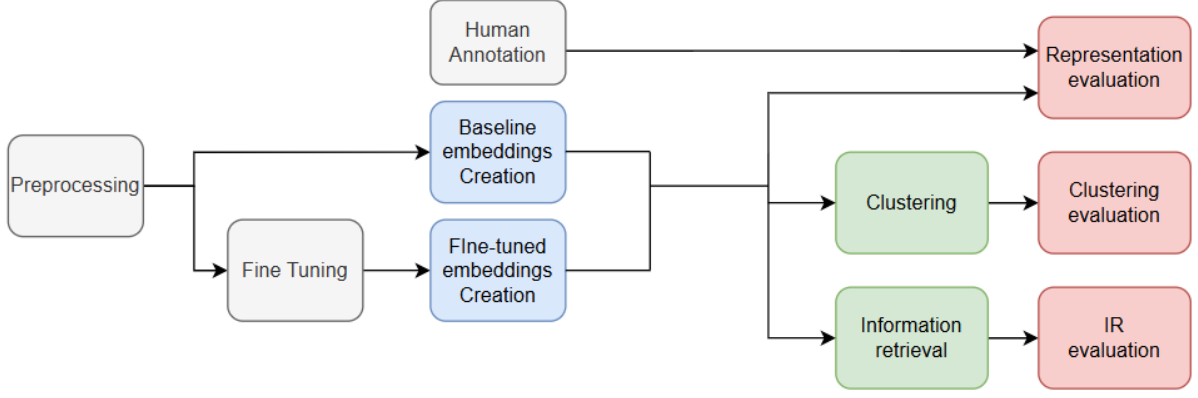


Figure 2: Pipeline

binary diagnostic (AUC) is derived by thresholding human similarity scores, and regression-style metrics such as MSE and MAE are reported to quantify absolute error. Finally, a paired t-test on model cosine similarities is performed to determine whether the fine-tuned embeddings are significantly closer to human judgments.

3.4 Clustering

To study global structure, we cluster document embeddings and assess both internal cohesion and alignment with MeSH-based topics.

Embeddings are first reduced via PCA to 50 components (for stability and noise reduction), then clustered with K-Means. The number of clusters is fixed ($k = 20$) based on standard silhouette/elbow diagnostics.

Internal indices (Silhouette and Davies–Bouldin) and external alignment are reported with MeSH using Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI). Cluster purity is computed by the fraction of documents in each cluster that share its majority MeSH label. For qualitative analysis, we surface representative documents nearest to each centroid (cosine distance), together with their MeSH terms, to verify topical coherence.

In the end a lightweight bootstrap (re-clustering repeated subsamples of the corpus) provides variability estimates for ARI/NMI/silhouette and enables paired significance tests between baseline and fine-tuned spaces.

3.5 Information retrieval

Finally, we evaluate semantic search using dense vectors only (no lexical retrieval or cross-encoder re-ranking).

- **Setup:** From the original test corpus, 500 medical abstracts were randomly sampled to serve as queries. Each query corresponds to a medical abstract annotated with a set of MeSH terms. Embedding representations for both query and candidate documents were computed using the models under evaluation (namely, the baseline and fine-tuned variant). These embeddings were used to generate ranked lists of documents for each query, which were subsequently evaluated using standard information retrieval metrics.

- **Ground Truth Construction:** The relevance between a query and a document was defined based on *MeSH-term* overlap. Specifically, a document was considered relevant to a query if it shared at least two terms with it. For each of the 500 sampled queries, we constructed a relevance set by comparing its MeSH annotations with those of all documents in the test collection. On average, each query was associated with X relevant documents, reflecting the overall sparsity of relevance across the dataset.

- **Ranking and metrics:** Queries and documents were ranked by cosine similarity between their embedding vectors, computed using either the baseline or fine-tuned model. No approximate indexing or re-ranking was applied, so performance reflects the quality of the embedding space. Retrieval effectiveness was measured using standard metrics: **Mean Reciprocal Rank (MRR)**, **Mean Average Precision (MAP)**, **Recall**, and **nDCG**, for $K \in \{1, 5, 10, 50, 500\}$.

4 Results

This section presents the outcomes obtained from the experiments described in the implementation, comparing the baseline SBERT model (all-MiniLM-L6-v2) with the fine-tuned version adapted on MeSH-based sentence pairs.

4.1 Data

4.1.1 Human-Aligned Similarity

Table 1 presents the correlation, discrimination, and error metrics between model-predicted cosine similarities and human-annotated scores across 300 abstract pairs. The fine-tuned encoder consistently yields higher correlation and lower error values.

Metric	Baseline	Fine-tuned	Δ
Pearson r	0.598	0.563	-0.036
Spearman ρ	0.637	0.599	-0.038
AUC (threshold ≥ 3)	0.780	0.750	-0.03
MSE	1.6925	1.7575	+0.06
MAE	1.1713	1.1965	+0.02
p -value (paired t -test)	—	-4.67	—

Table 1: Human-aligned similarity on 300 abstract pairs.

4.1.2 Clustering

Table 2 reports internal and external validation indices obtained by applying PCA (50 dimensions) followed by K-Means clustering ($k = 20$). Bootstrap averages over ten resamples are reported in Table 3.

Metric	Baseline	Fine-tuned	Δ
Silhouette	0.065977	0.083344	+0.018
Davies–Bouldin	2.865401	2.535741	-0.33
ARI (vs. MeSH)	2.052547e-07	3.628151e-07	-1.575e07
NMI (vs. MeSH)	0.441816	0.439636	-0.02
Purity	0.000624	0.000678	+0.00054

Table 2: Clustering quality with PCA (50D) followed by K-Means ($k = 20$).

Metric	Baseline	Fine-tuned	p -value
ARI	0.010	0.010	0.6582
NMI	0.4665	0.4670	0.4022
Silhouette	0.0608	0.0740	0.002

Table 3: Bootstrap robustness (10 subsamples, 70% each; MiniBatchKMeans). Mean and paired significance tests.

4.1.3 Information Retrieval

Table 4 summarizes retrieval metrics computed over 500 random queries, each evaluated against

documents sharing at least two MeSH terms with the query. Normalized discounted cumulative gain (nDCG) values at standard cutoffs are also presented.

Metric	Baseline	Fine-tuned	Δ
MRR	1	1	0
MAP	0.5029	0.5109	+0.08
Recall@1	0.0001	0.0001	0
Recall@5	0.0002	0.0002	0
Recall@10	0.0004	0.0004	0
Recall@50	0.0016	0.0017	+0.001
Recall@500	0.0149	0.0153	+0.0004
nDCG@1	1	1	0
nDCG@5	0.7493	0.7424	-0.0069
nDCG@10	0.6860	0.6837	-0.0023
nDCG@50	0.6027	0.6084	+0.0057
nDCG@500	0.5429	0.5546	+0.0117

Table 4: Dense retrieval over 500 queries (relevance defined as ≥ 2 shared MeSH terms).

4.2 Discussion

The results show that MeSH-guided domain adaptation affects different evaluation dimensions unevenly. For RQ1, the fine-tuned SBERT model does not improve similarity alignment with human judgments: correlations decrease slightly and error increases (table 1). This suggests that optimizing embeddings toward MeSH-based topical similarity shifts the cosine space in a way that reflects domain taxonomy rather than the semantic cues humans consider when rating sentence-level similarity.

For RQ2, however, the domain-adapted model produces notably stronger internal cluster structure. Silhouette improves (+0.018) and Davies–Bouldin decreases, indicating tighter and more coherent clusters (table 2). Bootstrap analysis confirms a statistically significant gain in Silhouette ($p = 0.002$), while external MeSH alignment remains largely unchanged—likely due to the mismatch between hierarchical multi-label MeSH and flat K-Means clustering. These findings suggest that fine-tuning primarily sharpens the global geometry of the embedding space rather than improving strict label alignment.

For RQ3, dense-only retrieval shows small but consistent gains at deeper ranks, whereas top-rank metrics remain stable or slightly worse (table 4). Since no lexical or cross-encoder components were used, these improvements can be attributed directly to representation learning. The strict relevance criterion (more than 2 MeSH overlaps) further constrains early-rank performance, explaining the limited difference at low cut-offs.

Across trials, the same qualitative pattern emerged: better cluster geometry and deeper-rank retrieval, with neutral or slightly negative effects on human-aligned similarity. This confirms that MeSH-based fine-tuning primarily enhances topical structure rather than fine-grained semantic similarity.

5 Conclusions

This study evaluated SBERT domain adaptation for clinical text across similarity assessment, clustering, and dense retrieval. Fine-tuning with MeSH-derived positive pairs sharpened the global embedding structure (improving internal clustering metrics and modestly enhancing deeper-rank retrieval) while providing no benefit for human-aligned sentence similarity. These outcomes reflect the nature of the supervision signal: MeSH emphasizes topical proximity rather than semantic granularity, shaping the resulting embedding space accordingly.

Because the experiments used dense-only retrieval with a strict MeSH-based relevance definition, the reported improvements represent a conservative estimate of real-world gains. Integrating the fine-tuned embeddings into a complete IR pipeline (lexical retrieval + cross-encoder re-ranking) is expected to amplify the benefits observed here.

Future work should explore improved supervision signals (e.g., UMLS-augmented pairs, harder negatives), clustering methods compatible with multi-label semantics, and broader evaluations on additional clinical corpora to enhance generalizability.

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